STUDENT MANAGEMENT DATABASE

Project Overview | Microsoft SQL Server

1. Project Description

This project is a relational database designed to organize information for an educational environment. It stores and connects data about students, instructors, courses, departments, and course topics. The database also records student enrollments and instructor-course assignments, helping keep related information structured and consistent.

2. Main Tables

Table	Purpose
Students	Stores student details such as name, age, and department reference.
Instructors	Stores instructor information, including name, contact/address details, salary-related fields, and department reference.
Courses	Stores course details such as course name, duration, description, and topic reference.
Departments	Stores department information and its related instructor reference.
Student_Courses	Associates students with courses and stores the grade for each enrollment.
Instructor_Courses	Associates instructors with courses and stores an evaluation value.
Topics	Stores course topics/categories used to classify courses.

3. Relationships Between Tables

• **Students and Departments:** Students are associated with departments through the department identifier.

• **Students and Courses:** Student_Courses acts as a junction table connecting students and courses. It also stores each student's grade for a course.

• **Instructors and Courses:** Instructor_Courses acts as a junction table connecting instructors and courses, with an evaluation field.

• **Courses and Topics:** Each course is linked to a topic through the topic identifier.

• **Departments and Instructors:** The diagram includes department and instructor references to represent their association.

4. Database Design Concepts

The design uses primary-key and foreign-key relationships to identify records and connect related tables. Junction tables are used for the student-course and instructor-course associations. This structure reduces unnecessary duplication and supports organized data retrieval through SQL queries.

5. Tools

Database Platform: Microsoft SQL Server
Database Tool: SQL Server Management Studio (SSMS)

Note: This overview is based on the database diagram provided. The exact constraints, column data types, and business rules should be verified against the SQL script or database.